

AGROSPEAR CROPCARE CO., LTD.
PLANT GROWTH REGULATORS (PGR) QUALITY CONTROL SYSTEM & DOSSIER

植物生长调节剂 (PGR) 生产全流程质量控制体系与佐证报告规程

Document Code: SOP-QC-PGR-2026
Effective Date: August 2026
Version: Rev 4.0 (Global Standard)
Compliance: FAO/WHO Specs & CIPAC Methods
Product Lines: SP, SL, SC, WDG, TB PGR Formulations

PGR ULTRA-HIGH PRECISION & BIO-SAFETY PROTOCOL: Plant Growth Regulators exhibit physiological biological activity at ppm (mg/L) thresholds. Overdosing induces severe phytotoxicity or physiological malformation. Active assay tolerance must strictly conform to $\pm 2.5\%$ – 5.0% . Epimer & optical isomer ratios (e.g., Paclobutrazol 2R,3R-isomer / Brassinolide 14-epibrassinolide vs. 24-epibrassinolide) must be resolved via Chiral HPLC-MS.

植物生长调节剂 (PGR) 超高精度与生物安全特别规程：植物生长调节剂在 ppm (mg/L) 微量级即发挥极强生理调控作用，浓度偏高易导致严重药害、畸形或绝产。有效成分含量允差控制极其严苛 ($\pm 2.5\%$ – 5.0%)。关键手性异构体与同分异构体比例（如多效唑 2R,3R-抑长活性体、芸苔素内酯 14-羟基与 24-表芸苔素内酯比例）必须通过手性高分辨液相色谱精准测定。

1. END-TO-END PGR QA/QC WORKFLOW / PGR 生产品控全生命周期流程

QC Stage / 阶段	Key Control Parameters & Protocols / 关键管控项目与规范	Methodology / 检验方法与仪器	Release Criteria / 准则
IQC (Incoming QC) 原辅料进厂检验	• Technical Material (TC): AI purity, stereoisomer ratios (e.g., Uniconazole / Paclobutrazol optical isomers), related fermentation impurities (for Gibberellic Acid GA3/GA4+7), moisture. • 原药(TC): 有效成分纯度、手性同分异构体比例（如烯效唑/多效唑活性光学异构体）、赤霉素发酵杂质（GA3/GA1/GA4+7 区分）及微量水分。 • Co-solvents & Stabilizers: Organic solvents (Ethanol/Isopropanol anhydrous grade), non-ionic wetting agents, effervescent disintegration couples (citric acid + sodium bicarbonate). • 助溶剂与崩解体系: 无水级溶剂含水量 < 0.2%、泡腾崩解对（食品级柠檬酸与碳酸氢钠干燥度及纯度）。 • Packaging: Moisture-impermeable aluminum-foil pouches with nitrogen purging for water-soluble powders/tablets (SP/TB), fluorinated barrier bottles for SL/SC. • 专用包材: 充氮铝箔复合防潮袋（SP 可溶粉/TB 泡腾片）、高阻隔氟化 HDPE 瓶（SL/SC）。	• Chiral HPLC / UHPLC-MS • Karl Fischer Coulometer • Effervescent Reaction Assay • Nitrogen Sealing Tester	100% Passed 不合格严禁入库
IPQC (In-Process) 制程过程巡检	• Soluble Powder / Granule (SP/SG): Micro-dosing coefficient of variation (CV < 1.5%), 3D rotary blending homogeneity, environmental humidity RH < 35% (preventing caking). • 可溶粉/可溶粒剂(SP/SG): 超微量成分混合变异系数 (CV < 1.5%)、三维混料均一性、除湿洁净车间环境湿度 RH < 35%。 • Soluble Liquid (SL): Complete solvent dissolution end-point, low-temp crystallization monitoring at -5°C, flash point safety validation. • 水剂/乳油(SL/EC): 原药完全溶解度、-5°C 极低温耐寒析晶监控与溶剂闪点测试。 • Effervescent Tablets (TB): Tablet hardness (≥ 45 N), friability (< 0.5%), dissolution bubbling time (< 90s). • 泡腾片剂(TB): 片剂硬度 (≥ 45 N)、脆碎度 (< 0.5%)、水中自分散气泡崩解时间 (< 90秒)。 • Filling Line: High-precision dosing ($\pm 0.2\%$ tolerance), induction foil hermetic seal, 2D QR serialization. • 灌装分装线: 高精度微量称重分装（误差 $\pm 0.2\%$）、铝箔热封闭性与一物一码追溯赋码。	• Homogeneity HPLC Sampling • Tablet Hardness / Friability • Digital Psychrometer • In-line Multi-head Weigher	CCP In-Spec 微量偏差立即停线
FQC / OQC 成品终检与放行	• Ultra-Trace Quantitative Assay: High-sensitivity HPLC/GC baseline quantification of active ingredients (e.g., 0.01% Brassinolide / 0.1% Forchlorfenuron CPPU). • 超微量有效成分精确测定: 高灵敏度 HPLC-DAD/MS 基线定量（如 0.01% 芸苔素内酯、0.1% 氯吡脞 CPPU 含量）。 • Physical Properties: Complete water solubility, Solution clarity & transparency (CIPAC MT 41), Foaming test, pH stability (CIPAC MT 75.3). • 理化特性: 水溶速度与透明澄明度、溶液热储沉淀、起泡量 ≤ 15 mL、pH 稳定性。 • Stress Degradation: Accelerated thermal stability (54$\pm 2^\circ\text{C}$, 14 days) & cold storage (0$\pm 2^\circ\text{C}$, 7 days) with AI degradation $\leq 3.0\%$. • 稳定性破坏试验: 54°C 热储 14 天有效成分分解率 $\leq 3.0\%$，0°C 冷储 7 天无沉淀、无分层。	• CIPAC MT 41 / MT 174 • CIPAC MT 46.3 / MT 39.3 • HPLC-MS/MS System	COA Released QA 总监签字放行
Traceability & Bio-assay 生物活性验证与追溯	• Biological verification: Standard crop hypocotyl elongation / mung bean rooting bio-assay verifying batch biological activity consistency. • 生物学活性验证: 标准作物下胚轴伸长/绿豆插条生根生物测定法，校验批次间生理活性一致性。 • Retained samples archived at 15–20°C for shelf-life + 1 year (min. 3 years) linked to digital ERP batch records. • 恒温留样库保存至保质期后 1 年，与 ERP 数字化批生产记录（BMR）及检验图谱绑定追溯。	• Growth Chamber Bioassay • Cloud ERP Traceability	Archived 建立终身追溯档案

2. FORMULATION-SPECIFIC QC STANDARDS (FAO/CIPAC COMPLIANT) / 主要植物生长调节剂剂型质控指标

Formulation / 剂型	Representative Products / 代表品种	Critical Quality Attributes (CQA) / 核心理化质控指标	Key Test Method / 对应方法
SL / SP (Soluble Solution/ Powder) 可溶性水剂 / 可溶粉	Gibberellic Acid (GA3) 40% SP / 4% SL Ethephon 480g/L / 720g/L SL Mepiquat Chloride 250g/L SL / 98% TC	Active Content: Nominal $\pm$ 5.0% Dissolution Time: $\leq$ 30 seconds (100% clear) pH Range: GA3 (3.0 – 6.5) / Ethephon ($\leq$ 3.0) Water Content: $\leq$ 1.5% (for SP)	HPLC / Titration CIPAC MT 41 CIPAC MT 75.3
Ultra-low Concentration Formulations 超微量制剂 (SL/EC/SP)	14-Hydroxylated Brassinolide 0.01% SL 24-Epibrassinolide 0.1% SP Forchlorfenuron (CPPU) 0.1% SL	Micro-assay Precision: High recovery rate $\geq$ 98% Active Isomer Ratio: Standard bioactive epimer Clarity & Transparent Appearance: Stable at 0°C Thermal Stability (54°C, 14d): AI degradation $\leq$ 3.0%	HPLC-MS/MS Chiral Separation CIPAC MT 46.3
SC / WDG (Suspension/ Granule) 生长延缓悬浮剂 / 水分散粒剂	Paclobutrazol 250g/L SC / 15% WP Uniconazole 5% SC / 5% WP Prohexadione-calcium 10% / 15% WDG	Active Enantiomer Ratio: 2R,3R-isomer $\geq$ 99.0% Particle Size: D50 $\leq$ 1.8μm, D90 $\leq$ 3.5μm (SC) Suspensibility: $\geq$ 92.0% (SC) / $\geq$ 88.0% (WDG) Residual Moisture: $\leq$ 1.5% (Prohexadione-Ca WDG)	Chiral HPLC Laser Particle Size CIPAC MT 184 / 174
TB (Effervescent Tablets) 泡腾片剂	Gibberellic Acid (GA3) 10% / 20% TB 6-BA (6-Benzylaminopurine) + GA4+7 TB	Effervescent Disintegration Time: $\leq$ 90 seconds Tablet Hardness: 45 – 80 N Friability: $\leq$ 0.50% (CIPAC MT 178) Weight Variation: Within $\pm$ 2.0%	CIPAC MT 174 Tablet Hardness Tester Electronic Balance

3. SUPPORTING EVIDENCE 1: OFFICIAL COA TEMPLATE / 佐证文件一：成品出厂检验报告单 (COA)

AGROSPEAR CROPCARE CO., LTD.

CERTIFICATE OF ANALYSIS / 产品检验分析报告单 (PGR)

Product Name / 品名: 14-Hydroxylated Brassinolide 0.01% SL (14-羟基天然芸苔素内酯 0.01% 水剂)

Batch No. / 生产批号: AS-PGR-20260826-15

Invoice / Order No. / 订单号: AGR-EXP-2026-978

Batch Quantity / 批量: 5,000 Liters (100mL x 50/Carton)

Manufacture Date / 生产日期: 2026-08-26

Expiry Date / 有效期至: 2028-08-25

Sampling Date / 取样日期: 2026-08-26

Inspection Date / 判定日期: 2026-08-27

Test Parameter / 检验项目	Specification Limit / 标准指标	Test Result / 实测结果	Standard Method / 检验方法	Verdict / 判定
Appearance / 外观性状	Clear yellowish transparent liquid (澄清淡黄色透明液体)	Conforms / 符合要求	Visual Check	Pass
Active Ingredient: 14-Hydroxylated Brassinolide	0.0100 $\pm$ 0.0005 % (w/w) (0.0095% – 0.0105%)	0.0102 % (w/w)	HPLC-MS/MS (Agilent 6470)	Pass
pH Value (1% aqueous solution)	4.5 – 7.5	5.8	CIPAC MT 75.3	Pass
Density (20°C) / 密度	0.980 – 1.020 g/cm³	1.005 g/cm³	CIPAC MT 3.2.1	Pass
Water Solubility & Solution Clarity	100% transparent, no haze, no precipitate	Clear / 完全澄清	CIPAC MT 41	Pass
Persistent Foaming / 持久起泡性	$\leq$ 15 mL after 1 min	5 mL	CIPAC MT 47.2	Pass
Low Temp. Stability (0 $\pm$ 2°C, 7 days) / 冷储	No precipitation, crystals or phase separation (无析出)	Conforms / 澄明无析出	CIPAC MT 39.3	Pass
Heat Stability (54 $\pm$ 2°C, 14 days) / 热储	AI Degradation $\leq$ 3.0%, Appearance unchanged	Degradation: 0.6%, Clear	CIPAC MT 46.3	Pass
Bio-activity Verification / 生物测定	Hypocotyl elongation rate $\geq$ 120% vs. blank control	Conforms (128.5%)	Standard Bioassay	Pass

CONCLUSION / 检验结论: PASS - Fully complies with FAO Specifications & In-house Quality Standard. Approved for release. (合格 - 各项微量分析与理化指标均符合 FAO 规格及企业内控标准, 准予放行出厂。)

Analyst / 检测员:
Zhang Wei (张伟)
Date: 2026-08-27

Reviewer / 复核员:
Li Ming (李明)
Date: 2026-08-27

QA Director / QA总监:
Dr. Alex Chen (陈博士) [Approved]
Date: 2026-08-27

4. SUPPORTING EVIDENCE 2: BATCH RELEASE & OOS PROTOCOL / 佐证文件二：批放行审批与超标处理规程

Release Gate / 质量关卡	Verification Requirements / 审核项目	Status / 状态	Verified By / 审核人
1. Raw Material Gate (IQC)	Active ingredient purity & bioactive isomer confirmation, anhydrous solvent moisture < 0.2%, nitrogen packaging seal	Verified Pass	IQC Lead
2. Process Parameters (IPQC)	Micro-dosing blending homogeneity (CV < 1.5%), workshop humidity RH < 35%, TB tablet hardness & bubbling speed	Verified Pass	IPQC Officer
3. Laboratory FQC (COA)	High-sensitivity HPLC-MS/MS trace assay, water solubility clarity, cold/thermal stability, hypocotyl bioassay	Verified Pass	Lab Supervisor
4. Packaging & Traceability	2D QR code serialization, high-precision automated filling weight verification, hermetic seal test	Verified Pass	Packaging QA

PGR OOS & HIGH-SENSITIVITY DISPOSITION PROTOCOL: Due to extreme physiological sensitivity, any active ingredient deviation exceeding $\pm 5.0\%$ or failure in bioactive isomer ratio triggers immediate batch quarantine. Reworking is restricted and requires a strict three-party technical review signed off by the Chief Formulation Chemist, Factory General Manager, and QA Director.

植物生长调节剂超标（OOS）与处置规程： 鉴于调节剂极高生理敏感度，任何有效成分含量偏差超过 $\pm 5.0\%$ 或活性异构体比例异常，系统自动锁定出库闸口并张贴物理红卡。返工调配受到严格受控限制，必须经由首席配方科学家、工厂总经理与质量总监三方联合技术审查签署后方可执行。